

HDOC Day_29 Report Programming in C

Problem

Write a C program to perform the following menu-driven operations:

1. Count even numbers, odd numbers, positive, negative, and zero elements in an array.
2. Search for an element in an array using linear search.
3. Merge 2 arrays.
4. Exit

C Program Code

```
#include <stdio.h>

int main()
{
    int ch;

    printf("choice 1 is to Count even numbers, odd numbers, positive, negative, and zero elements in an array.\n");

    printf("choice 2 is to search for an element in an array using linear search.\n");

    printf("choice 3 is to combine 2 arrays\n");

    printf("exit is choice 4\n");

    printf("enter your choice\n");

    scanf("%d",&ch);

    switch (ch)
    {
        case 1:
        {
            int n;int even=0;

            int odd=0;
```

```

int positive=0;
int negative=0;
int zero=0;
printf("enter the size of array\n");
scanf("%d",&n);
int arr[n];
for(int i=0;i<n;i++)
{
    scanf("%d",&arr[i]);
}
for(int i=0;i<n;i++)
{
    if(arr[i]%2==0)
        even++;
    else
        odd++;
    if(arr[i]>0)
        positive++;
    else if(arr[i]==0)
        zero++;
    else
        negative++;
}

printf("the number of positive elements are %d,the number of negative elements
are %d\nnumber of even elements are %d,number of odd elements are %d,\nand
number of zeros are %d",positive,negative,even,odd,zero);

break;
}

case 2:

```

```

{
    int n;

    printf("enter the size of array\n");

    scanf("%d",&n);

    int arr[n];

    for(int i=0;i<n;i++)

    {

        scanf("%d",&arr[i]);

    }

    int num;

    printf("enter the element to be searched\n");

    scanf("%d",&num);

    for(int i=0;i<n;i++)

    {

        if(arr[i]==num)

        {

            printf("element found at %d\n",(i+1));

        }

    }

    break;

}

case 3:

{

    int n;

    printf("enter the size of array\n");

    scanf("%d",&n);

    int arr1[n];

    printf("array 1\n");

```

```
for(int i=0;i<n;i++)
{
    scanf("%d",&arr1[i]);
}

int m;

printf("enter the size of array\n");

scanf("%d",&m);

int arr2[m];

printf("array 2\n");

for(int i=0;i<m;i++)
{
    scanf("%d",&arr2[i]);
}

int z=m+n;

int arr3[z];

for(int i=0;i<z;i++)
{
    if(i<m)
        arr3[i]=arr2[i];
    else
        arr3[i]=arr1[i-m];
}

for(int i=0;i<z;i++)
{
    printf("%d\n",arr3[i]);
}

break;
}
```

```
case 4:
{
    return 0;
}
default:
{
    printf("invalid choice\n");
    break;
}
}
return 0;
}
```